

Multispectral satellite imagery for illegal waste material classification

State of the art and thesis direction

Alessandro Potenza, M.Sc. Computer Science & Engineering · AI

The problem: priority, not just presence

Illegal waste dumping is an environmental crime and a public-health issue.

Agencies such as ARPA cannot inspect everything, they must prioritise by material.

Today's automated pipelines find sites well; they classify presence, not what is there.

Research question

What is the added value of multispectral data, vs. RGB only, for waste material classification from satellite imagery?

Why material matters: $\text{risk} = \text{hazard} \times \text{exposure} \times \text{magnitude}$

- The hazard is material-bound: inert rubble vs plastics vs asbestos-cement are worlds apart.
- Material carries the hazard; the European Waste Catalogue flags hazardous codes with * (asbestos 17 06 05*).
- Priority risk = hazard (material) x exposure (who is nearby) x magnitude (how much).
- So identifying the MATERIAL - not just locating the site - is the decision-relevant question.

Today's paradigm: RGB deep learning on aerial imagery

VHR aerial / satellite tile (RGB) → CNN or Transformer backbone → pretraining (ImageNet, EO SSL) → two-step fine-tune → binary waste / no-waste.

- Accurate within the geographic context it was trained on.
- Classifies presence, not material.
- Tied to chromatic information only.
- Generalisation collapses outside it, new region or new sensor.

These limits motivate looking at the spectrum.

State of the art - RGB deep learning detects sites, not materials

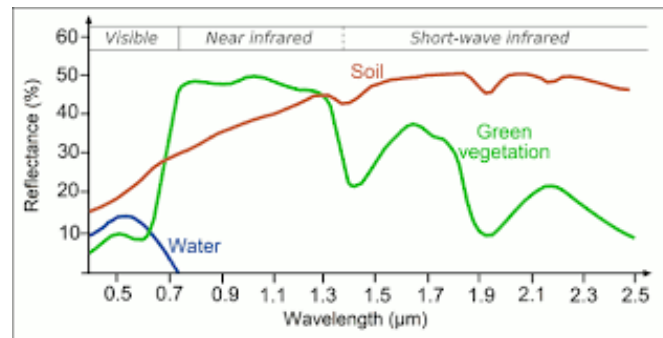
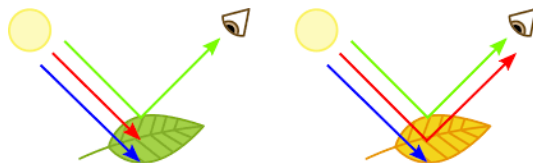
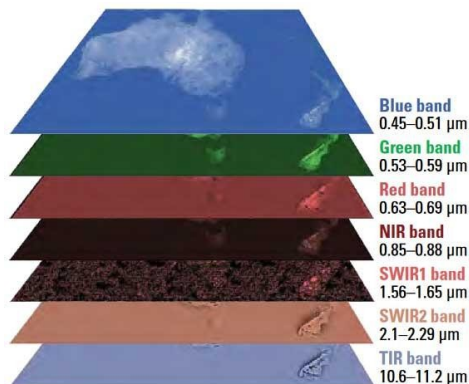
Work (year)	Input / GSD	Method	Key result
Gibellini 2025	AerialWaste RGB 20 cm	Swin-T + RSP, two-step	F1 92.0%; cross-region -5.1%
AerialWaste (Torres 2023)	aerial RGB 20-50 cm	dataset; ResNet+FPN	F1 80.7%; 22 cats, RGB-only
Sun 2023 (global)	VHR RGB 0.3-1 m	BCA-Net detection	~2,500 dumpsites; sens. 98%
CascadeDumpNet 2024	Pleiades 0.5 m RGB	cascade CNN + AutoML	mAP 84.6%; transfers across cities
Disaitek 2024	Pleiades Neo 0.3 m	operational service	waste ≥ 2 m ² ~95% (vendor)

A satellite pixel is a spectrum, not just a colour

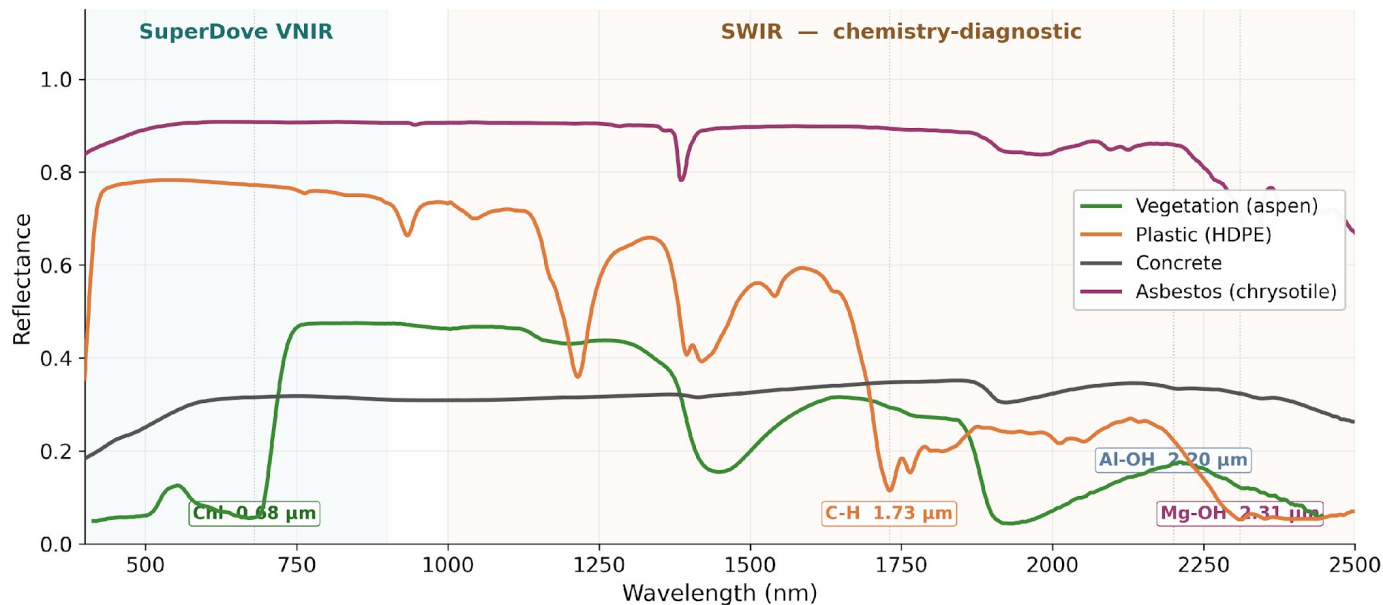
Each pixel is a vector of reflectance values, the material's spectral signature.

- RGB, 3 broad visible bands (colour only).
- Multispectral (MS), 4-15 chosen bands (e.g. WorldView-3 8 VNIR + 8 SWIR; Pleiades Neo 6 VNIR).
- Hyperspectral (HSI), hundreds of contiguous narrow bands (EnMAP 230; USGS splib07a 2,151 ch).

This vector is the raw input from which any classifier reasons.



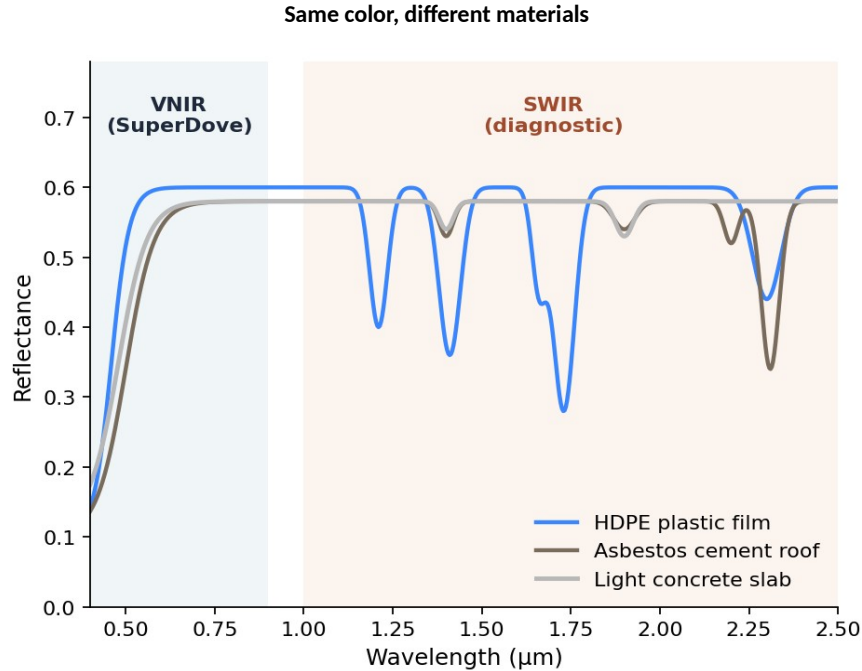
Every material has a spectral fingerprint



USGS splib07a (Kokaly et al., 2017)

RGB fails in two distinct ways (Iso-chromaticity)

iso-chromaticity and sub-pixel mixing, both invisible without spectral information.



HDPE plastic film

Asbestos-cement roof

Light concrete slab

Where each hazard becomes separable: feature -> band

Where each hazard becomes separable: diagnostic feature to band

	Pleiades Neo (VNIR-only)			WorldView-3 adds SWIR			
Asbestos (Mg-OH)						weak	strong
Plastics (C-H)				strong	strong	weak	weak
Concrete/C&D (carbonate)						strong	strong
Slag/rust (Fe-oxide)		weak	strong				
AC weathering (moss/lichen)		strong	weak				
	RGB	Red Edge	NIR	SWIR 1.2um	SWIR 1.73um	SWIR 2.2um	SWIR 2.33um

USGS splib07a · Aguilar 2021/2025 · Cilia 2015

State of the art - spectral material discrimination: asbestos

Work (year)	Input / GSD	Method	Key result
Shepherd 2025	EnMAP HSI 30 m	8-classifier cascade (ACE)	86% field match; OA 91.4%
Cilia 2015	MIVIS airborne HSI 3 m	SAM + MNF, ISD index	PA 89% / UA 86%
Saba 2026 †	WV-3 VNIR 1.24 m	32 classifiers, Fine-KNN	Macro-F1 97.6% (VNIR only)
Bonifazi 2026 †	WV-3 16 b 1.24/3.7 m	Max-Likelihood + aggregation	AC roofs; multi-temporal monitoring
Abbasi 2024 †	aerial RGB (no SWIR)	DenseNet + LSTM, temporal	OA ~96%, AC 94% by shape+time

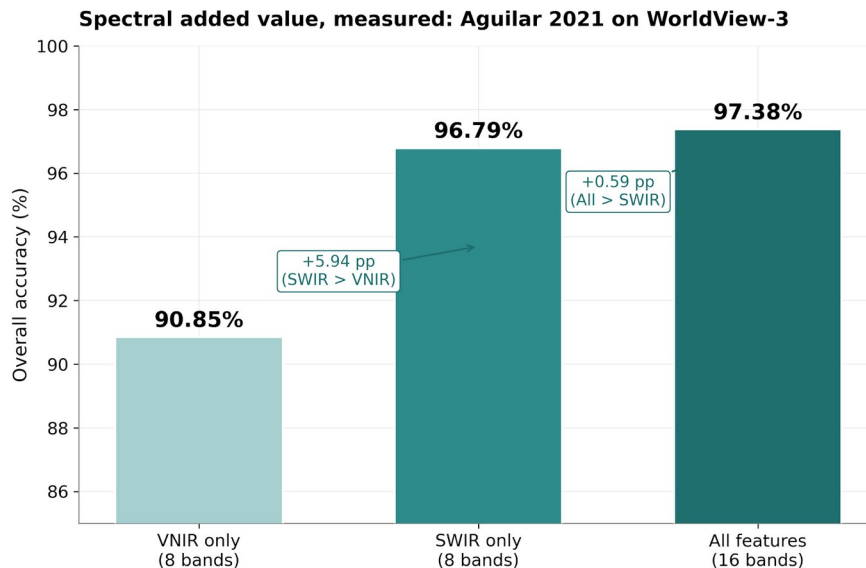
State of the art - spectral material discrimination: plastics & urban materials

Work (year)	Input / GSD	Method	Key result
Aguilar 2021	WV-3 VNIR + SWIR	OBIA + DT, 3-way ablation	OA 90.85 -> 96.79 -> 97.38
Aguilar 2025	WV-3 SWIR 3.7 m	matched filter (USGS spectra)	precision 92.5%; lab-image r=0.95
EMIT (Estrela 2025)	EMIT HSI 60 m	matched filter, HDPE/PVC	first global orbital plastic map
MARIDA 2022	Sentinel-2 10 m	RF / U-Net benchmark	F1 0.79; sub-pixel mixing at 10 m
CDW 2025	lab HSI 768 b	narrowband + MLP	RGB 0.87 -> +2 NIR 0.96 ~ full HSI
SpectralWaste 2024	RGB + SWIR HSI (conveyor)	RGB/HSI/fusion segmentation	fusion mIoU 58 > RGB 48; HSI wins thin classes

Spectral added value, measured: Aguilar 2021 on WorldView-3

Most-cited band ablation on WV-3 for waste-like classification (greenhouse plastic, $n = 14.3$ M pixels).

OBIA + Decision Tree, 10-fold CV. The canonical evidence that SWIR adds material-discriminative power.

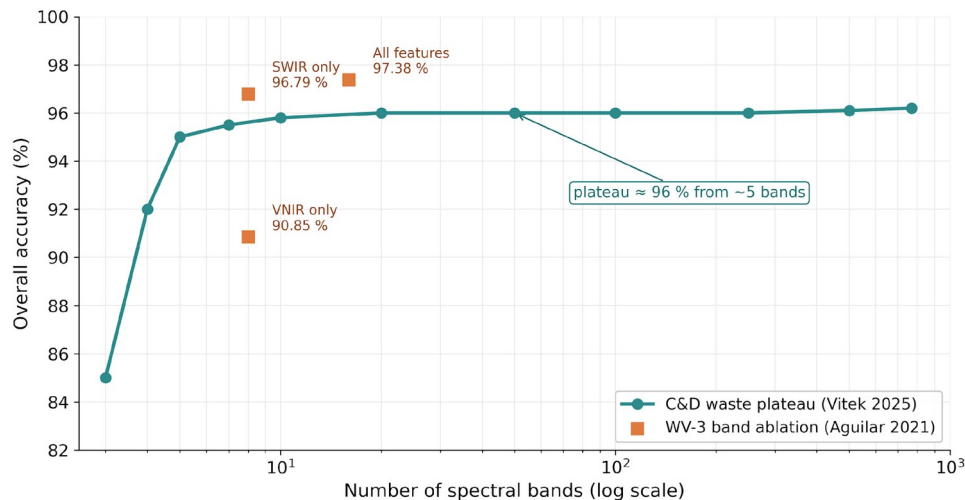


Aguilar, Jiménez-Lao & Aguilar 2021 — Remote Sensing 13(11):2133. OBIA + Decision Tree, 10-fold CV, 14.3 M pixel GT.

More bands $\neq$ more information, well-chosen few suffice

- Aguilar 2021 (WV-3 plastic): VNIR 90.85 $\rightarrow$ All 97.38; SWIR carries the jump.
- CDW 2025 (lab HSI, C&D): RGB 0.87 $\rightarrow$ RGB + 2 NIR bands 0.96, on par with full 768-band HSI.
- Zhou 2021 (WV-3 SWIR): 8 narrow bands separate aliphatic vs aromatic polymers.

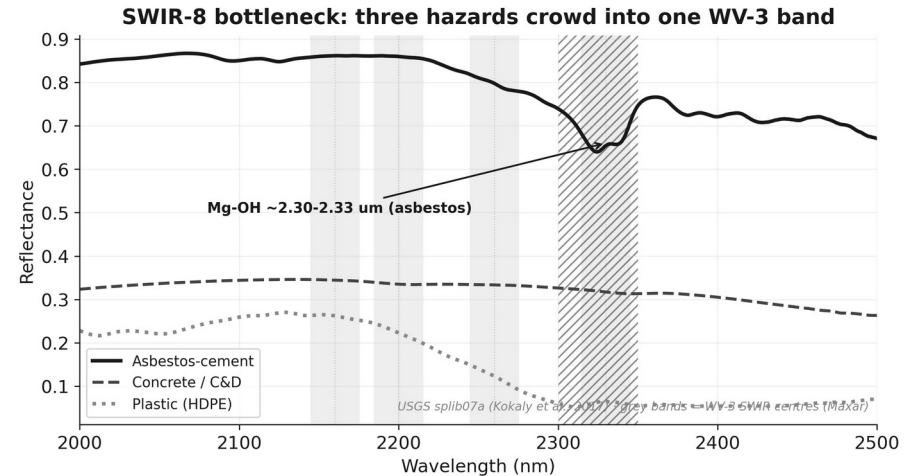
What matters is which bands, not how many: a few well-chosen SWIR bands recover most of the gain.



Vitek et al. 2025 (CTU Prague, C&D waste) — Stage-1/2 narrowband selection on HSI 425–975 nm · Aguilar 2021 verified above.

Honest limit: resolution is split between texture and chemistry

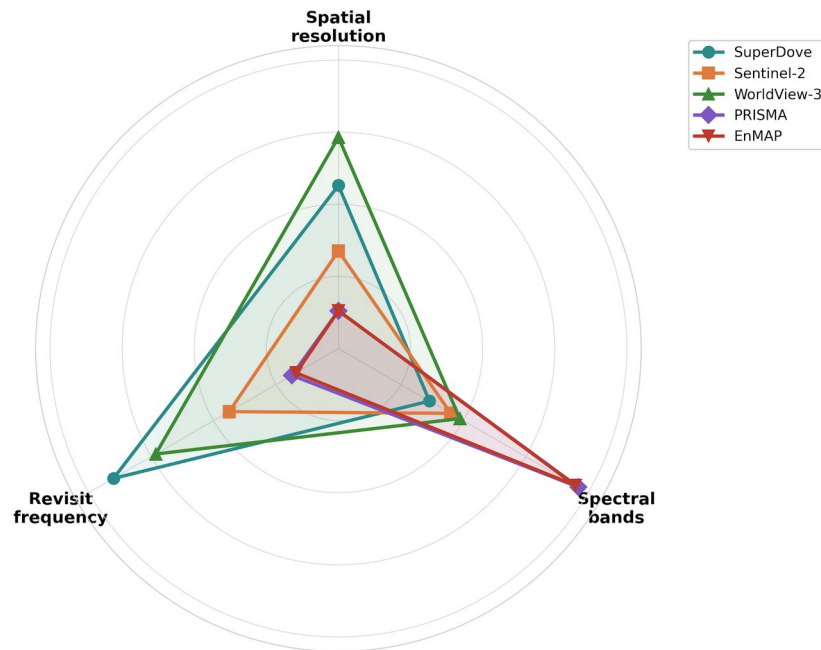
- Spectra are sampled coarsely: VNIR ~1.24 m, SWIR ~3.7 m (~14 m² pixels).
- The pan band gives fine texture ~0.3 m, but no material info alone.
- A small dump is easier to SEE (texture) than to chemically IDENTIFY (spectra).
- SWIR-8 bottleneck: asbestos 2.32 + concrete 2.34 + plastic 2.31 crowd one band.
- And SWIR is not always needed: Saba 2026 / Abbasi 2024 classify asbestos at VHR with none.



The sensor trade-off: spatial x spectral x revisit

Three axes, one photon budget. No satellite maximises all three.

- WorldView-3: 1.24 m VNIR + 3.7 m SWIR, 16 bands - the only VHR + SWIR option.
- Pleiades Neo: 1.2 m, 6 VNIR, no SWIR - sharpest VNIR-only.
- Sentinel-2 / EnMAP: 10-30 m - rich spectra, too coarse for material at site scale.
- SuperDove: 3 m, 8 VNIR, free, near-daily - wide-area but VNIR-only.



Outer ring = better. Axes: Spatial 1/GSD (log) · Bands log₁₀(N) · Revisit 1/days (log).
Specs: Planet, ESA, Maxar, ASI, DLR (mission documents).

Very-high resolution: the SWIR divide

	WorldView-3	Pleiades Neo	SuperDove
VNIR	8 b @ 1.24 m	6 b @ 1.2 m	8 b @ 3 m
SWIR	8 b @ 3.7 m	none	none
Pan	~0.31 m	~0.30 m	-
Access	commercial / TPM	commercial / TPM	free, near-daily

WorldView-3 is the only sub-2 m platform that also carries SWIR - where the chemistry lives.

Pleiades Neo and SuperDove are VNIR-only: sharp, but blind to the diagnostic SWIR absorptions.

The trade-off the whole SOTA turns on: spatial detail vs spectral (chemical) information.

State of the art - datasets and the structural data gap

Work (year)	Input / GSD	Method	Key result
AerialWaste (Torres 2023)	aerial RGB 20-50 cm	22 material categories	RGB-only; coordinates withheld
CWLD 2024	GF-2 0.8 m + GE 0.5 m	C&D segmentation masks	F1 88.9% / IoU 82% (China)
MARIDA 2022	Sentinel-2 10 m	15-class pixel benchmark	marine debris - water, not land
SpectralWaste 2024	RGB + SWIR HSI	conveyor-belt segmentation	sorting plant, not Earth observation

Foundation models for Earth Observation (EO): a new lever

2024–2025: a wave of pretrained backbones for EO.

Most are pretrained at 10–30 m on Sentinel-2 / HLS, naïve transfer to 3 m SuperDove is not guaranteed.

Two routes forward

- Sensor-agnostic models (DOFA, AnySat), accept arbitrary band sets.
- Parameter-efficient adapters (DEFLECT), freeze the backbone, add small heads (<1 % parameters).

Open question: does pretrain at 10–30 m transfer well to 3 m?

State of the art - foundation models and N-band adaptation

Work (year)	Input / GSD	Method	Key result
DOFA (Xiong 2024)	5 modalities 1-30 m	wavelength hypernetwork	up to 202 bands; new-sensor plug-in
AnySat (Astruc 2025)	11 sensors 0.2-250 m	JEPA multimodal	SOTA on 9 tasks
Prithvi-EO-2.0 2024	HLS 6 b incl. SWIR 30 m	3D temporal MAE	+8 pp GEO-Bench; fixed bands
SoftCon 2024	Sentinel-1/2 13 b	soft contrastive + DINOv2	SOTA 10/11; random-init extra bands
DEFLECT 2025	MS 6-13 b	PEFT adapter <1% params	on par with full fine-tune

DOFA: a band-agnostic backbone, the candidate to start from

Dynamic One-For-All (Xiong et al. 2024).

- A hypernetwork generates patch-embedding weights from each band's central wavelength.
- One backbone ingests any sensor (WV-3 16 b, Pleiades Neo 6 b, Sentinel-2 13 b, HSI) by wavelength.
- Makes an RGB vs VNIR vs SWIR comparison fair - the substrate for a controlled band ablation.
- Pairable with PEFT adapters (DEFLECT, <1% params) for cheap spectral fine-tuning.

State of the art - the object-vs-material boundary

Work (year)	Input / GSD	Method	Key result
Ramachandran 2024	VHR sub-metre	DL object detection	tanks P 0.96 / R 0.97; >169k mapped
YOLOv7-OT 2024	VHR satellite	YOLOv7 + CBAM	tanks 90% acc / 95.9% precision
ELV Hybrid-YOLOv5 2025	close-range infrared	scrap-metal detection	mAP 84%; needs spectrum / close-range
UAV solid-waste 2024	UAV	dual-branch segmentation	OA >94%; generic pile, no per-material

Gaps the state of the art reveals

- RGB ceiling: every operational VHR waste detector is RGB-only and learns shape/context, not material.
- No public dataset combines VHR + terrestrial waste + material labels (AerialWaste is RGB; CWLD is C&D-only).
- Material evidence is adjacent, not direct: proven on greenhouses / asbestos roofs in isolation, never in a waste pipeline.
- Generalization (cross-region / cross-sensor / cross-time) is fragile and unmeasured for MS.
- Foundation models pretrain at 10-30 m and are untested for material at VHR.
- No link from material -> hazard class -> risk, the decision output agencies actually need.

Generalization: fragile and largely unmeasured for MS

	In-domain	Cross-region	Cross-sensor
RGB (today)	strong	-5.1% F1 (Gibellini)	ports trivially
+ VNIR	-	gap narrows (physics)	band-shift fragile
+ SWIR	-	gap narrows most	widens unless harmonized

A recurring SOTA blind spot: most works report in-domain only. Whether added bands narrow or widen the gap is itself an open question.

What the state of the art points to

The open question is sharp and unanswered: how much does multiband beat RGB for waste MATERIAL - and where does it stop?

- A controlled band ablation (RGB -> +RedEdge/NIR -> full VNIR -> +SWIR) on a wavelength-agnostic backbone (DOFA).
- Generalization as a first-class axis (cross-region / sensor / time), not an afterthought.
- Candidate VHR platforms span the SWIR divide: WorldView-3 (with SWIR) vs Pleiades Neo (VNIR-only).
- Asbestos is the cleanest first material: public ground truth + a textbook SWIR signature.

Independent of any single dataset: the SOTA defines the experiment, not the data in hand.